

Ex.No:1: Simulation of FDMA in MATLAB Date

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

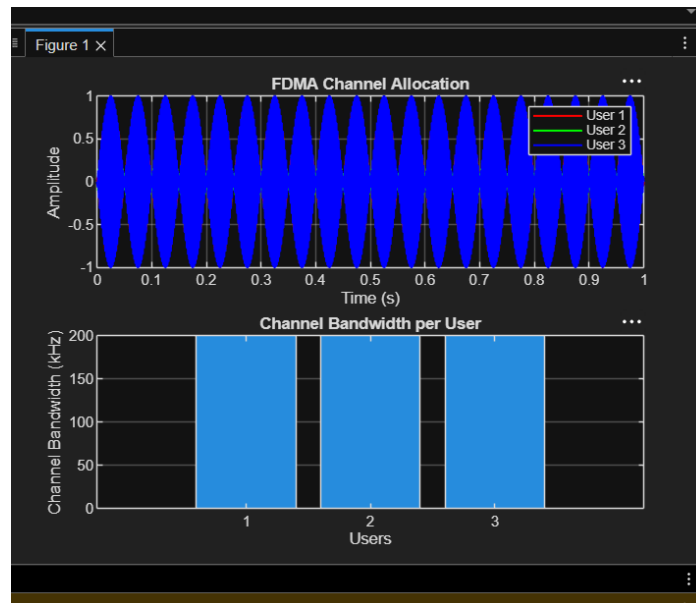
Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

Program (Matlab Code)

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;
CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time
(s)'),ylabel('Amplitude') subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

Output:



Output Graph 1: FDMA Channel Allocation

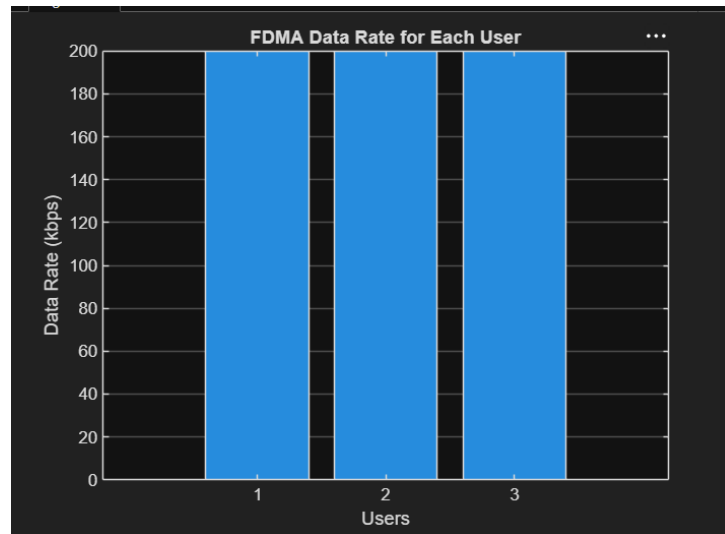
- Displays three users transmitting on different carrier frequencies (100 kHz, 300 kHz, and 500 kHz).
- Demonstrates that each user occupies a separate frequency channel.

Output Graph 2: Channel Bandwidth Allocation

- Shows a bar graph of the bandwidth allocated to each user.
- Each user is assigned 200 kHz, confirming equal FDMA channel allocation.

Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

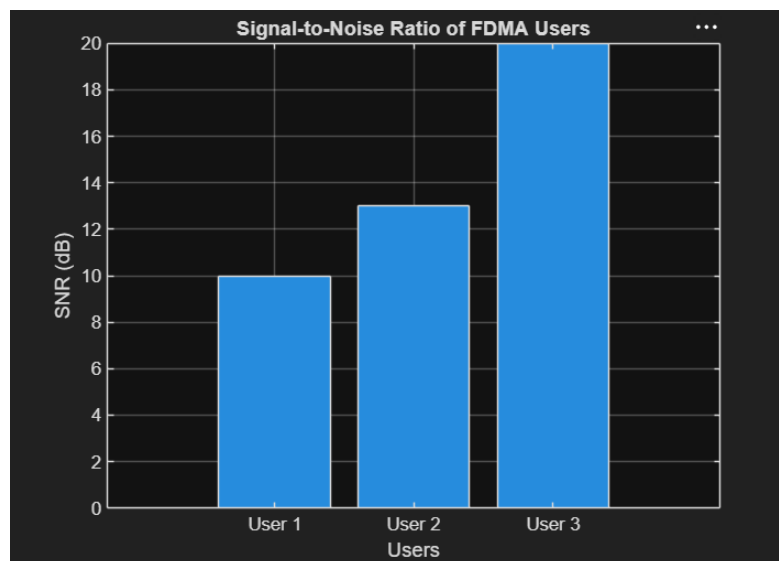


The graph shows that all FDMA users achieve an equal data rate of 200 kbps due to equal channel bandwidth allocation. This confirms fair and interference-free data transmission among users.

Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to-Noise Ratio (SNR) for different users.

OUTPUT



The graph shows that the SNR increases as the noise power decreases for different FDMA users. A higher SNR indicates better communication quality and more reliable signal transmission.